

SHELBYVILLE, IN, USA

WMO: 724356

Lat: 39.578N

Lon: 85.803W

Elev: 804

StdP: 14.27

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53866

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.9	8.9	-5.8	4.1	4.9	-0.5	5.5	10.0	26.9	34.1	24.7	34.2	6.7	300	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	91.3	74.4	89.2	73.5	87.1	72.5	77.8	87.2	76.4	85.2	75.0	83.3	9.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	134.6	83.6	73.3	127.7	81.7	72.3	123.3	80.3	41.7	87.3	40.2	85.3	38.9	83.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.4	18.3	-5.3	93.9	8.3	3.8	-11.3	96.6	-16.2	98.8	-20.8	100.9	-26.9	103.7		
			-6.1	80.3	7.9	1.5	-11.8	81.3	-16.4	82.2	-20.9	83.0	-26.6	84.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	29.1	32.0	43.1	54.7	64.9	73.0	74.9	73.5	67.7	55.7	43.9	34.3	(6)
(7)		DBStd	18.65	11.95	11.58	11.17	9.33	8.61	5.70	5.33	4.99	7.16	9.36	9.45	10.53	(7)
(8)		HDD50	2276	654	514	276	58	4	0	0	0	46	229	495		(8)
(9)		HDD65	5128	1114	926	682	328	114	8	1	3	52	317	632	951	(9)
(10)		CDD50	3736	5	9	62	198	466	689	771	727	531	222	47	9	(10)
(11)		CDD65	1114	0	0	3	17	110	248	307	266	133	29	1	0	(11)
(12)		CDH74	9920	0	1	26	167	962	2118	2850	2294	1248	248	6	0	(12)
(13)		CDH80	3388	0	0	3	18	288	745	1066	781	427	60	0	0	(13)
(14)	Wind	WSAvg	8.4	10.4	10.0	10.1	10.5	8.8	7.4	5.8	5.2	6.0	8.1	9.2	9.6	(14)
(15)	Precipitation	PrecAvg	43.6	3.0	2.5	3.3	4.3	5.1	5.0	4.3	3.5	3.1	3.0	3.4	3.1	(15)
(16)		PrecMax	55.6	9.1	5.2	6.5	8.5	8.8	10.9	8.5	6.0	7.4	7.0	7.4	6.5	(16)
(17)		PrecMin	33.5	0.9	0.9	0.9	1.6	1.4	1.5	1.8	1.1	0.1	0.6	1.1	0.6	(17)
(18)		PrecStd	6.7	1.9	1.3	1.5	1.8	2.0	2.3	1.8	1.3	2.0	1.7	1.5	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.6	66.9	77.8	82.5	89.6	92.5	95.8	93.2	92.5	86.2	72.8	63.6	(19)
(20)		0.4%	MCWB	58.1	57.1	62.9	64.0	71.3	74.0	77.1	74.9	71.8	69.3	60.4	57.7	(20)
(21)		2%	DB	56.7	59.4	71.3	79.1	86.2	89.9	91.3	89.8	88.6	80.5	67.2	58.6	(21)
(22)		2%	MCWB	53.3	53.6	59.9	62.5	68.7	72.9	75.9	74.5	71.6	66.2	58.4	54.4	(22)
(23)		5%	DB	52.1	54.1	65.0	75.0	83.5	87.6	88.7	87.3	84.9	75.9	63.2	53.9	(23)
(24)		5%	MCWB	49.2	49.3	56.5	61.1	68.3	72.0	74.8	73.8	69.6	63.5	56.5	50.4	(24)
(25)		10%	DB	46.3	48.4	61.1	70.7	79.9	84.5	86.0	84.3	81.4	71.8	59.1	49.5	(25)
(26)		10%	MCWB	42.7	43.3	54.0	58.8	66.7	70.5	73.5	72.5	68.0	61.9	52.7	46.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.6	59.2	64.3	68.5	73.4	77.8	80.3	79.3	75.8	72.0	63.9	59.4	(27)
(28)		0.4%	MCDWB	61.8	65.0	76.1	78.7	83.8	87.7	90.1	88.6	85.7	82.0	69.5	62.3	(28)
(29)		2%	WB	53.9	54.6	61.1	65.4	71.7	76.0	78.2	77.4	74.3	69.2	60.4	54.8	(29)
(30)		2%	MCDWB	56.4	58.5	68.8	73.7	82.8	85.5	87.8	86.5	84.2	77.5	65.0	57.8	(30)
(31)		5%	WB	49.4	49.8	58.3	63.0	70.2	74.5	76.8	75.8	72.5	66.3	57.3	50.5	(31)
(32)		5%	MCDWB	52.0	54.1	63.8	70.7	80.2	83.6	85.7	83.7	81.4	72.9	62.4	53.3	(32)
(33)		10%	WB	42.8	43.7	54.5	60.6	68.5	72.9	75.3	74.4	70.6	63.1	53.4	46.1	(33)
(34)		10%	MCDWB	46.2	48.5	60.2	69.0	77.0	81.5	83.6	81.6	77.7	70.1	58.5	48.7	(34)
(35)	Mean Daily Temperature Range		MDBR	14.8	15.7	18.1	20.8	19.3	19.0	19.2	20.1	22.2	20.5	17.6	13.7	(35)
(36)		5% DB	MCDBR	18.7	21.3	23.3	25.4	23.0	22.3	21.2	22.5	26.1	24.9	22.6	18.1	(36)
(37)		5% DB	MCWBR	15.9	17.0	14.6	13.5	10.0	9.4	9.1	10.6	11.2	12.8	15.6	15.7	(37)
(38)		5% WB	MCDBR	17.5	20.0	20.5	19.4	19.4	18.3	18.6	18.9	20.7	19.7	19.0	16.9	(38)
(39)		5% WB	MCWBR	16.4	17.4	13.7	12.7	9.5	9.1	9.2	10.0	9.8	11.1	15.0	15.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.329	0.351	0.406	0.445	0.449	0.460	0.455	0.417	0.360	0.323	0.304	(40)	
(41)		taud	2.429	2.337	2.359	2.213	2.152	2.187	2.156	2.157	2.244	2.405	2.457	2.449	(41)	
(42)		Ebn at Noon	274	281	287	276	266	265	260	258	261	267	265	266	(42)	
(43)		Edh at Noon	26	33	36	44	48	46	47	46	40	30	25	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	561	813	1151	1516	1721	1952	1898	1739	1430	1011	678	467	(44)	
(45)		RadStd	57	95	104	124	150	117	114	96	123	96	75	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+122

Nomenclature: See separate page